The Islamic University of Gaza

Faculty of Information Technology

Department of Computer Science

Fall Semester 2025–2026
Programming 1 Lab
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Online Store System – Final Project .

After the customer logs in, the system displays the **Customer Menu**.
This menu allows the customer to interact with the store, view products, and manage the shopping cart.

Customer Menu :

```
1. Browse All Products
2. Add Product to Cart
3. View My Cart
4. Logout
```

The menu:

- Is displayed repeatedly using a loop
- Accepts only valid numeric choice

How Data Is Stored (Arrays Design)

The store inventory is stored using three parallel arrays of size 100 :

Store Products Arrays.

1- One array for product names
2- One array for product prices
3- One array for product quantities

Important Rule:

The same index in all arrays represents **one product**.

productNames = [“book”,-----]

productPrices = [100,-----]

productQuantities = [5,-----]

Shopping Cart Array :

The shopping cart is stored using one array.

Cart Data Structure

- The system uses **one array for cart quantities**.
- Each index in the cart array represents **one product added to the cart**.

Important Note:
When the user selects a product, the quantity of that product is stored in the **cart quantity array at the same index** where the product appears in:

- the product names array
- the product quantities arra

Option 1: Browse All Products

When the customer selects Browse All Products:

- The system displays all products in the store
- Each product shows:
    - Product number (index + 1)
    - Product name
    - Product price
    - Product quantity

Sold Out Products

- If a product's quantity is 0, the system displays "SOLD OUT"
- Sold out products cannot be added to the cart

This helps the customer understand product availability clearly.

Option 2: Add Product to Cart

The customer adds a product by choosing its number from the displayed product list.

Product Selection by Number

- The number corresponds to the product's position in the product arrays

Validation Checks

1- Product Exists

2- Requested Quantity Is Available

Successful Addition to Cart:

If all validations pass:

1- The requested quantity is saved in the cart

2- The product quantity in the store inventory is **reduced**

3- A success message is displayed

Option 3: View My Cart

When the customer selects **View My Cart**:

- All cart items are displayed along with total price
- Each item shows:
  - Product name
  - Quantity in the cart

If the cart is empty:

- The system displays a clear message .

Cart Options

Inside the cart menu, the customer can:

1. Remove a product from the cart
2. Update product quantity
3. Return to customer menu
4. Proceed to checkout

All choices are validated to prevent invalid actions.

Remove Product from Cart

When removing a product:

- The product is removed from the cart arrays
- The quantity removed is returned to the store inventory

Key Logic:
The store and the cart must always stay synchronized.

Update Product Quantity in Cart

The customer can change the quantity of a cart item.

Invalid Quantity

- Zero or negative quantities are rejected
- The system shows an error message

Proceed to Checkout

When the customer proceeds to checkout:

- The system confirms the purchase(Total price)
- A success message is displayed
- The system is End

This simulates a completed purchase.

Validation Rules :

Throughout the system:

1- All menu inputs are validated

2- All quantities are checked

3- Clear error messages guide the user

Important Notes

1- Clear and meaningful variable names should be used.

2- The program must handle invalid inputs gracefully (no crashes).

3- All menus must loop correctly until the user chooses to exit.

4- The project can be completed by a maximum of 3 students per group.

5- Each group member should understand every part of the code.

6- You Can apply Object-Oriented Programming (OOP) concepts to complete this project.

7- It is recommended to separate each main operation into a static method.

8- Final submission deadline: 20 / 2 / 2026

9- Any copy-paste or plagiarism is strictly forbidden.

10- The use of Artificial Intelligence tools (including ChatGPT) to generate code or project solutions is strictly prohibited.

11- The project must be submitted as one PDF file only. The PDF must start with the complete Java source code added by copy–paste (no screenshots of code). After the code, the PDF must include clear screenshots for all required cases and scenarios (such as browsing products, sold-out product, adding to cart, exceeding quantity, viewing cart, updating/removing items, and checkout). For each screenshot, a short explanation is mandatory, describing what action was taken, what input was entered, and why the system produced that result.

12- only one student from the group must submit the complete project

13- Project discussions will start on 21 / 2 / 2026.

Best Wishes 😎 🎩

Aya Nabil Alharazin

5 Feb 2026